

JEE MAIN REVISION MASTER: ENVIRONMENTAL CHEMISTRY OUT OF SYLLABUS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 96 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Environmental Chemistry — Core Principles & Governing Laws	68 PYQs	Core Principle
Very High	Environmental Chemistry — Applied Cases, Reactions & Traps	23 PYQs	Applied Topics
Moderate	Environmental Chemistry — Standard Calculations & Problem Archetypes	7 PYQs	Quantitative Analysis

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Environmental Chemistry — Core Principles & Governing Laws	NO_3^- , ClONO_2
Environmental Chemistry — Applied Cases, Reactions & Traps	$\text{Cl(g)} + \text{O}_3\text{(g)} \rightarrow \text{ClO(g)} + \text{O}_2\text{(g)}$, $\text{CF}_2\text{Cl}_2\text{(g)} \xrightarrow{\text{uv}} \text{Cl(g)} + \text{CF}_2\text{Cl(g)}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Environmental Chemistry — Core Principles & Governing Laws (Very High Yield — 68 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Environmental Chemistry.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Environmental Chemistry.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Environmental Chemistry PYQs.

Representative Source PYQs:

- [2023 Morning Shift] Q#1 (MCQ): "The possibility of photochemical smog formation will be minimum at ..." [Ans: N/A]
- [2023 Morning Shift] Q#1 (MCQ): "How can photochemical smog be controlled?..." [Ans: N/A]
- [2023 Evening Shift] Q#1 (MCQ): "The concentration of dissolved Oxygen in water for growth of fish should be more than X ppm and Biochemical Oxygen Demand in..." [Ans: N/A]

3.2 Environmental Chemistry — Applied Cases, Reactions & Traps (Very High Yield — 23 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Environmental Chemistry.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2023 Evening Shift] Q#1 (MCQ): "Which of the following compounds is an example of Freon?..." [Ans: N/A]
- [2023 Evening Shift] Q#1 (MCQ): "Which of these reactions is not a part of breakdown of ozone in stratosphere?..." [Ans: N/A]
- [2023 Morning Shift] Q#2 (MCQ): "Given below are two statements: Statement I : If BOD is 4 ppm and dissolved oxygen is 8 ppm, then it is a good quality water. Statement II : ..." [Ans: N/A]

3.3 Environmental Chemistry — Standard Calculations & Problem Archetypes (Moderate Yield — 7 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Environmental Chemistry.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2023 Morning Shift] Q#2 (MCQ): "Given below are two statements: Statement I : If BOD is 4 ppm and dissolved oxygen is 8 ppm, then it is a good quality water. Statement II : ..." [Ans: N/A]
- [2022 Evening Shift] Q#2 (MCQ): "Given below are two statements : One is labelled as Assertion A and the other is labelled as Reason R. Assertion A : Polluted water may have..." [Ans: N/A]
- [2022 Morning Shift] Q#3 (MCQ): "Given below are two statements: Statement I : In polluted water values of both dissolved oxygen and BOD are very low. Statement II : Eutro..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2023 (Online) 15th April Morning Shift (Q#1)

The possibility of photochemical smog formation will be minimum at :

- (A) Mumbai in May
- (B) Kolkata in October
- (C) New-Delhi in August (Summer)
- (D) Srinagar, Jammu and Kashmir in January

Correct Answer: None

Solution Analysis: The formation of photochemical smog depends on several factors, including sunlight, temperature, and the presence of pollutants such as nitrogen oxides and volatile organic compounds. Option (A) Mumbai in May may also have some possibility of photochemical smog formation because the temperature and sunlight are relatively high in May. Option (B) Kolkata in October may have some possibility of photochemical smog formation because the temperature and sunlight are still relatively high in October. Option (C) New-Delhi in August (Summer) is more likely to have photochemical smog formation because

Benchmark Problem 2: JEE Main 2023 (Online) 11th April Evening Shift (Q#1)

Which of the following compounds is an example of Freon?

- (A) $C_2H_2F_2$
- (B) C_2HF_3
- (C) C_2F_4
- (D) $C_2Cl_2F_2$

Correct Answer: None

Solution Analysis: Freons are a class of compounds which are also known as chlorofluorocarbons (CFCs). As the name suggests, these compounds contain carbon, chlorine, and fluorine atoms. So the compound that is an example of a Freon from the options given is : Option D : $C_2Cl_2F_2$ This compound contains both chlorine and fluorine atoms bonded to carbon, and is therefore a CFC or a Freon.

Benchmark Problem 3: JEE Main 2023 (Online) 11th April Morning Shift (Q#2)

Given below are two statements: Statement I : If BOD is 4 ppm and dissolved oxygen is 8 ppm, then it is a good quality water. Statement II : If the concentration of zinc and nitrate salts are 5ppm each, then it can be a good quality water. In the light of the above statements, choose the most appropriate answer from the options given below :

- (A) Statement I is incorrect but Statement II is correct
- (B) Both the statements I and II are incorrect
- (C) Both the statements I and II are correct
- (D) Statement I is correct but Statement II is incorrect

Correct Answer: None

Solution Analysis: Statement I : If BOD is 4 ppm and dissolved oxygen is 8 ppm, then it is a good quality water. Statement II : If the concentration of zinc and nitrate salts are 5 ppm each, then it can be a good quality water. The Biochemical Oxygen Demand (BOD) measures the quantity of oxygen used by microorganisms in the oxidation of organic matter. Lower values of BOD indicate better water quality. A BOD value of less than 5 ppm indicates good quality water. Therefore, in Statement I, the BOD value of 4 ppm indicates good water quality. Dissolved oxygen (DO) refers to the level of free, non-compound oxygen p